Subramanya Nagabhushanaradhya

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EDUCATION

University of Massachusetts — Amherst, MA
Master of Science in Computer Science — Sept 2021 - May 2023
Coursework: Neural Networks, Advanced Natural Language Processing, Intelligent Visual Computing, Mathematical Statistics, Systems for Data Science, Reinforcement Learning

The National Institute of Engineering — Mysore, India
Bachelor of Engineering in Computer Science — Aug 2015 - May 2019
Coursework: Operating System, Data Structures & Algorithms, Data mining and warehousing

SKILLS

Programming: Python, C++, JavaScript, SQL, Shell Scripting
ML/AI: PyTorch, LLMs, Agentic Workflows, Fine-grained Contextual Late Interaction
Security & Identity: OAuth 2.0, OIDC, Enterprise AI Security, API Gateway Security
Tools: Git, Linux, three.js, Unreal Engine

EXPERIENCE

Dylog Inc (dylog.ai) – AI agents for wholesale distribution — *San Jose, CA*
Founding Machine Learning Engineer & Chief Architect, Agentic Commerce — Aug 2023 - July 2026

- Architected Dylog's agentic commerce platform: an AI layer that unifies a distributor's ERP, PIM, CRM, and product content into one permission-aware knowledge base, and powers agents that let customers find products, get answers, request quotes, and place orders in whatever channel they already use.
- Built the search and retrieval foundation behind every agent, combining semantic and keyword retrieval, neural reranking, and LLM query understanding, grounded in approved sources with human handoff when needed; improved search relevance 35% and lifted conversion 28% in A/B tests.

Agentic Labs Inc. – UC Berkeley SkyDeck, Batch 21 — *Berkeley, CA*
Co-Founder — June 2025 - Present

- Co-founded Agentic Labs in UC Berkeley SkyDeck's Batch 21 accelerator cohort ($200K investment); responsible for technical vision and core platform development.

Human-Centered Robotics Lab, UMass — *Amherst, MA*
Graduate Student Researcher (Advisor: Prof. Hao Zhang) — Sept 2022 - May 2023

- Experimenting with multi-modal learning approaches based on DETR and GLIP to obtain attribute-based representations from natural language descriptions and visual features for object instance recognition and tracking.

Ribbon Communications — *Westford, MA*
Machine Learning Engineer, Intern — Sept 2022 - Dec 2022

- Implemented a multi-modal deep learning model for categorizing of user behavior based on VoIP traffic, images, and textual data gathered from user browsing sessions. Worked end-to-end on the project, from data collection to successful deployment.

Vertica — *Pittsburgh, PA*
Software Engineer Intern — June 2022 - August 2022

- Worked with Data Optimization and Backup team to develop an algorithm to improve and optimize database re-shading method in Eon mode. Extended this algorithm to improve backtracking of newly created storage containers while performing operations like re-shard, ILM and alter table.
- Designed and developed pipelines for data analytic and machine learning teams to work on backup and restoring information.

Meta — *Amherst, MA*
Graduate Student Researcher (Advisors: Jay-Yoon Lee, Shane Moon, Andrea Madotto) — Jan 2022 - July 2022

- Proposed an advanced cross-modal transformer approach with a video frame-level contrastive loss to reveal the correlation between language queries and video clips, focusing on Natural Language Query tasks for the Ego4d dataset. This approach resulted in an improvement in the baseline with a **IoU 0.5 recall @ 5 of 8.2** and **recall @ 3 of 4.8**

RenderPub *Bangalore, India*
Head of Product Development Jun 2018 - May 2021

- Lead architect, developer, and designer for RenderPub Studio, Stitch & Walk, architectural visualization softwares.
- Developed Common Unreal Modules (CUM) - a modular C++ / Blueprint system for RenderPub's Unreal Engine based applications. CUM comprises modules like RenderPub Walk Exporter, Dynamic Sky and weather systems, run time 3D file importer, terrain system with sculpting tools at runtime, Mass Placement, Array Placement and Mesh Painting tools.
- Developed and implemented a 360-degree panoramic image loading algorithm utilizing tile-based streaming, three.js, WebGL, and OpenCV to facilitate seamless transitions between hotspots and accurate 3D projections of images.
- Developed a desktop application utilizing Next.js, Electron, and React to launch and auto-update the RenderPub suite of applications and projects, with features such as authentication, news feeds, accounts, profile settings, and more.

PUBLICATIONS, STANDARDS, AND RECOGNITION

Identity Management for Agentic AI – OpenID Foundation white paper (arXiv:2510.25819) Oct 2025

- Honored as the **first acknowledged contributor** among 30+ global identity experts in the OpenID Foundation's foundational white paper on AI agent security. The OpenID Foundation's standards (e.g., OAuth 2.0, OpenID Connect) are used by billions of people worldwide.

OpenID Connect for Agents (OIDC-A) 1.0 (arXiv:2509.25974 | Specification | WorkOS Feature) Apr 2025

- Authored technical specification extending OpenID Connect Core 1.0 for LLM-based agent authentication and authorization within OAuth 2.0 ecosystem.
- Featured in WorkOS official blog "Identity for AI: Who Are Your Agents and What Can They Do?" and highlighted at Identiverse 2025, the premier digital identity industry conference.

Guest Lecture for Government of India Oct 2025

- Invited by the Principal Chief Personnel Officer of Central Railway (95,000 employees) to deliver a guest lecture on "Using AI Tools Securely at Work to Boost Productivity" at their Mumbai headquarters.

Identerati Office Hours, Ep. 134: "OIDC-A: Securing AI Agent Identity" (Video) Aug 2025

- Featured guest on Gluu's identity and access management show, presenting the OIDC-A specification alongside Tobin South (Stanford).

Navigating UMass Amherst: A Handbook for International Students (Handbook) 2023

- Authored comprehensive handbook sharing insights and advice for international students navigating academic and cultural transitions at UMass Amherst.

WRITING AND OPEN SOURCE

Writing (subramanya.ai): Git Was Built for 2,500 Merges a Day, The Filesystem Is the Database, Break In, Break Out: AI Agent Security in 2026
Open source (github.com/subramanya1997): markdownfs (Rust virtual filesystem for agents), rnp (NumPy's core rewritten in Rust), Import-3D-models-ingame (runtime 3D importer for Unreal Engine)